

SECTION 4.0

DENTAL SURGERY

SURGEON

Stephen A. Schendel, MD, DDS, FACS

ANESTHESIOLOGIST

Richard A. Jaffe, MD, PhD

TEMPOROMANDIBULAR JOINT ARTHROSCOPY/ARTHROTOMY/ARTHROPLASTY

SURGICAL CONSIDERATIONS

Description

Temporomandibular joint (TMJ) surgical procedures include both open and closed surgical techniques.

TMJ arthrotomy involves a preauricular, postauricular, or endaural incision to gain access to the joint compartment. It usually is performed for severe fibrous adhesion removal in the TMJ, bony or fibrous ankylosis, tumor resection, chronic dislocation, painful nonreducing disc dislocation, and severe osteoarthritis. **Open TMJ** surgery may range from discoplasty, discectomy, arthroplasty, and/or eminoplasty (reshaping of articular eminencia) to optimize the fit of the disc, condyle, and fossa to total joint replacement utilizing costochondral grafts or total joint metal implants. For the open treatment of condylar fractures, extraoral approaches (e.g., preauricular, retromandibular, and submandibular) are used. All extraoral approaches to the TMJ have the risks of facial nerve damage and the creation of visible scars. Due to those possible complications, endoscopically assisted transoral approaches for open reduction and miniplate fixation of condylar mandible fractures are being used more often. In any of these procedures, facial nerve monitoring is indicated.

TMJ arthroscopy is a minimally invasive technique that has reduced the need for open surgery of the TMJ. Arthroscopic TMJ surgery is indicated for treatment of internal derangements and intracapsular disorders. The major advantage is that it results in less periarticular tissue disruption and better preservation of vascular supply and lymphatic drainage of the joint. The procedure involves insertion of a TMJ miniscope through a preauricular puncture on the canthus-tragus line and insertion of an outflow needle. The joint compartment is continually lavaged with LR. A second cannula may be inserted. Arthroscopic TMJ procedures include lysis of adhesions and lavage, partial synovectomy, and abrasion arthroplasty. Sometimes a holmium:YAG laser is used to make intra-articular incisions anterior to displaced discs and to treat inflamed synovial tissue. Usually, at the end of the procedure, 2-mg dexamethasone is injected into the joint space. Injection of 2 mL 0.5% bupivacaine mixed with 1 mL sterile saline solution has been shown to significantly reduce postop pain. Arrhythmia, reflex bradycardia, and pulmonary edema have been reported as general complications in TMJ arthroscopy.

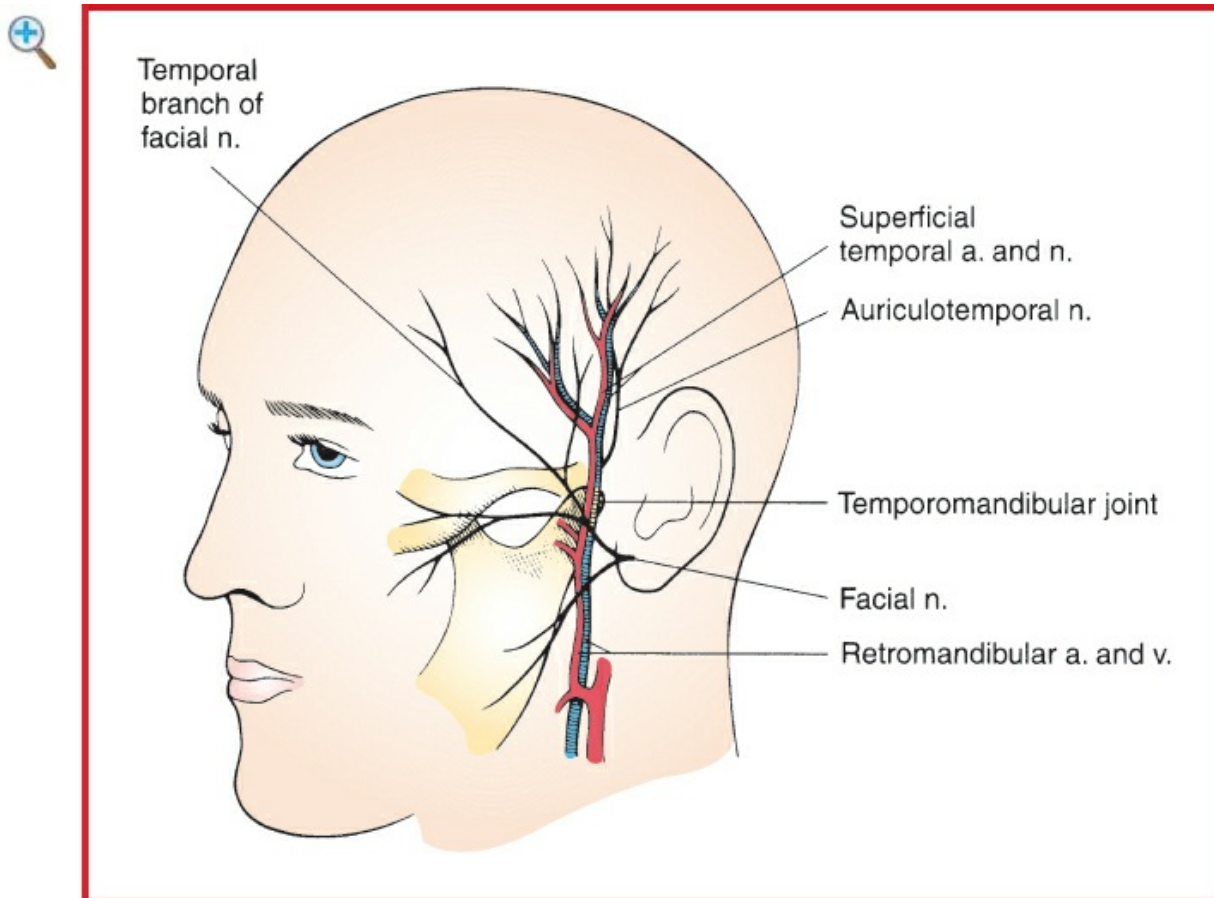


Figure 4.1 Anatomy for TMJ procedure.

Usual preop diagnosis

Internal derangement; subluxation; and ankylosis of TMJ

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Arthroscopy	Arthrotomy/Arthroplasty
Position	Supine	⇐
Incision	Preauricular	⇐
Special instrumentation	Arthroscope; laser	Power tools, endoscope, implant plates
Antibiotics	Cefazolin 2 g	⇐

Surgical time	0.5 h	1.5–3.5 h/side
EBL	Minimal	Minimal-moderate
Postop care	Outpatient procedure	24-h stay
Mortality	Minimal	⇐
Morbidity	VII nerve damage	⇐
	V nerve damage	⇐
	Hemorrhage	⇐
	Partial hearing loss	
	Ear fullness	
	Vertigo	
Pain score	5	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–40 yr
Male:Female	1:9
Incidence	20% of adult population suffers from TMJ dysfunction (TMJD)
Etiology	TMJD possibly 2° muscle spasm, bruxism, and osteoarthritis; idiopathic; trauma
Associated conditions	Psychiatric problems (typically depression); trismus; pain on opening the mouth; stress

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Dental/Oral Surgery.

Suggested Readings

1. Al-Ani Z, Gray R: TMD current concepts: 1. An update. *Dent Update* 2007; 34(5):278–80, 282–4, 287–8.
2. Furst IM, Kryshchak B, Weinberg S: The use of intra-articular opioids and bupivacaine for analgesia following temporomandibular joint arthroscopy: a prospective, randomized trial. *J Oral Maxillofac Surg* 2001; 59(9):979–83.
3. Hoffman D, Puig L: Complications of TMJ surgery. *Oral Maxillofac Surg Clin North Am* 2015; 27(1):109–24.
4. Indresano AT: Surgical arthroscopy as the preferred treatment for internal

- derangements of the temporomandibular joint. J Oral Maxillofac Surg 2001; 59(3):308–12.
5. Laskin DM: Temporomandibular disorders: the past, present and future. Odontology 2007; 95(1):10–5.
 6. Lieblich S: Providing anesthesia in the Oral and Maxillofacial Surgery Office: a look back, where we are now and a look ahead. J Oral Maxillofac Surg 2018; 76(5):917–25.
 7. Rajapakse S, Ahmed N, Sidebottom AJ: Current thinking about the management of dysfunction of the temporomandibular joint: a review. Br J Oral Maxillofac Surg 2017; 55(4):351–6.
 8. Rigon M, Pereira LM, Bortoluzzi MC, et al: Arthroscopy for temporomandibular disorders. Cochrane Database Syst Rev 2011; (5):CD006385.
 9. Tsuyama M, Kondoh T, Seto K, et al: Complications of temporomandibular joint arthroscopy: a retrospective analysis of 301 lysis and lavage procedures performed using the triangulation technique. J Oral Maxillofac Surg 2000; 58(5):500–5.

ORAL SURGERY

SURGICAL CONSIDERATIONS

Description

The most common surgeries of the oral cavity are third-molar removal, surgical extractions, apicoectomies, orthodontic exposures of teeth, osseointegrated implants, bone grafting, treatment of oral pathologic conditions, and preprosthetic surgery. **Surgical extractions** of teeth involve intraoral exposure of the roots through a mucosal incision and removal of the overlying bone with a surgical drill. Risks associated with removal of teeth in the mandible are damage to the inferior alveolar nerve (anesthetic numb lip), lingual nerve (anesthetic numb tongue), and, rarely, mandibular fracture. In the posterior maxilla, oroantral fistulas can occur and are closed with a mucoperiosteal flap. **Exposure of teeth** for orthodontic therapy involves creation of a mucoperiosteal flap and attachment of a bracket with a small gold chain on which the orthodontist can pull to integrate the tooth into the dental arch. **Grafting** to the maxilla and mandible is done for augmentation of the atrophied alveolar ridge and the maxillary sinus and in cases of cleft lip and palate, and can consist of bone and/or other materials to induce bone formation. A second team usually harvests the bone at the same time. Possible extraoral harvesting sites include the anterior or posterior iliac crest, the tibia, and the skull. **Preprosthetic surgery** of the oral soft tissue in preparation for dentures has been replaced largely by insertion of **osseointegrated implants** for retention of individual

teeth and dentures. Surgical treatment of **oral pathology** can range from removal of dentigerous cysts, with and without bone graft, to laser or surgical removal of mucosal lesions.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES				
	Dental Surgery	Dental Implants	Oral Pathology	Grafting
Position	Supine	⇐	⇐	Supine or prone
Incision	Intraoral	⇐	⇐	Intraoral and donor site
Special instrumentation	Surgical drill	Implant drill and kit	Surgical drill, laser	Special graft materials
Antibiotics	None	Penicillin 1 g	Cefazolin 2 g iv	⇐
Unique considerations	Nasotracheal intubation ±Throat pack	⇐ ⇐	⇐ ⇐	⇐ ⇐
Surgical time	0.5 h/tooth	0.5 h/implant	1–3 h	2–3 h
EBL	Minimal	⇐	⇐	Moderate
Postop care	Outpatient	⇐	⇐ or 24-h stay	24-h stay
Mortality	Minimal	⇐	⇐	⇐
Morbidity	V nerve damage Aspiration of dental debris	⇐ ⇐	⇐	Hemorrhage
Pain score	3	2	3–5	5

■ PATIENT POPULATION CHARACTERISTICS				
Age range	12–40 yr	>16 yr	All ages	>8 yr
Male:Female	1:1	⇐	⇐	⇐
Etiology	Idiopathic	Tooth loss	Various	⇐
Associated conditions	Craniofacial syndromes			

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Dental/Oral Surgery.

Suggested Readings

1. Bataineh AB: Sensory nerve impairment following mandibular third molar surgery. J Oral Maxillofac Surg 2001; 59(9):1012–7.
2. Zou L, He D, Ellis E: A comparison of clinical follow-up of different total temporomandibular joint replacement prostheses: a systematic review and meta-analysis. J Oral Maxillofac Surg 2018; 76(2):294–303.

Suggested Viewing

Links are available online to the following video:

TMJ Anatomy “The Joint” Video: <https://www.youtube.com/watch?v=nH8KG3fIoBM>

RESTORATIVE DENTISTRY

SURGICAL CONSIDERATIONS

Description

Multiple dental restorative procedures are performed under GA when there are rampant caries, and an extensive amount of dental work must be performed at one time. The second most common indication for GA is for procedures that need to be performed on mentally retarded patients who are not candidates for a local anesthetic. The actual amount of restorative dentistry is quite variable, depending on the individual case; thus, surgical time can be quite variable. Generally, blood loss is not a problem.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES

Position	Supine
Incision	Intraoral
Special instrumentation	Dental armamentarium
Unique considerations	Nasal intubation; throat pack

Antibiotics	Penicillin × 5 d po
Surgical time	0.5–3 h
EBL	Minimal
Postop care	PACU → home
Mortality	Minimal
Morbidity	Pain Aspiration of dental debris Swelling
Pain score	1–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	2 yr to adult
Male:Female	1:1
Incidence	Unknown
Etiology	Idiopathic or congenital anomalies
Associated conditions	Mental retardation (majority), Down syndrome, seizures

ANESTHETIC CONSIDERATIONS FOR DENTAL/ORAL SURGERY

PREOPERATIVE

Most patients presenting for dental or oral surgery usually will require only local anesthesia provided by the dentist/oral surgeon. Deep sedation or GA may be required, however, for several unique patient groups: (a) young children (some with systemic diseases such as CHD and hemophilia), (b) the mentally disabled, (c) those with poorly controlled seizure disorders, (d) those presenting for TMJ procedures, and (e) those with an oral septic focus, who may be quite ill. If the patient does not fall into one of these readily identifiable categories, the reasons for GA should be ascertained. An LMA with a flexible wire-reinforced airway tube (LMA Flexible) has been used successfully for a variety of oral surgical procedures. The use of an LMA should be discussed with the surgeon in advance because its presence may interfere with the planned procedure.

Airway	Patients presenting for TMJ procedures may have problems with mouth opening (2° pain, trismus, and arthritis), making airway examination difficult. Mouth opening may not improve with GA and muscle relaxation. Nasotracheal intubation using FOL (done awake in patients with difficult airways) should be planned. Examine nares for patency; check for loose teeth.
Respiratory	Surgery should be postponed (at least 2 wk) in patients presenting with Sx of acute RTI (fever, coughing, purulent sputum, etc.). Sx of chronic respiratory disease should be sought and treated before surgery. LMA use has been reported to decrease respiratory complications in children with upper RTIs. Tests: As indicated from H&P
Cardiovascular	Patients with dysrhythmias may be sensitive to the epinephrine used in local anesthetic solutions administered intraop. As with other types of elective surgery, preexisting cardiovascular problems should be treated before surgery. Prophylactic antibiotics for endocarditis are not required in most patients; exceptions include patients with prosthetic valves, congenital heart disease, h/o infective endocarditis, or heart transplant. Endocarditis prophylaxis: amoxicillin 2 g po or cefazolin 2 g iv 30–60 min before procedure. Tests: As indicated from H&P
Neurologic	Patients with seizure disorders should be on optimal medical therapy before surgery. Discuss precipitating factors and prodromal Sx with the patient. Tests: ✓ therapeutic levels of anticonvulsant (e.g., levetiracetam = 12–46 mcg/mL; phenytoin = 10–20 mcg/mL; carbamazepine = 3–12 mcg/mL; phenobarbital = 10–40 mcg/mL)
Musculoskeletal	In addition to TMJ problems, rheumatoid arthritis is

	associated with cricoarytenoid joint immobility and cervical spine immobility/instability that may complicate intubation. Patients with prosthetic joints do not require special antibiotic prophylaxis. Other tests as indicated from H&P
Laboratory	
Premedication	Standard premedication usually is appreciated, although, in patients with limited airway access, sedation may be inappropriate. If FOL is planned, pretreatment with an antisialagogue (e.g., glycopyrrolate 4 mcg/kg) is useful. Metoclopramide (e.g., 10–20 mg iv adult) will reduce the incidence of PONV 2° swallowed blood. Avoid long-lasting sedative medications.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen (adult: 1000 mg po; peds: 10–15 mg/kg iv or po as an elixir).

INTRAOPERATIVE

Anesthetic technique

GETA (although LMAs have been used successfully for many procedures). Typically a nasotracheal intubation is requested, using an ETT 0.5–1 mm smaller than for oral intubation. In patients with difficult airways, an awake nasal FOL is indicated. For conscious sedation, dexmedetomidine (e.g., 0.5 mcg/kg loading dose over 10 min + 0.5 mcg/kg/h infusion) in combination with midazolam (e.g., 0.03 mg/kg) ± propofol (e.g., 25 mcg/kg/min) has the advantage of minimal respiratory depression.

Induction	In patients with normal airways, a standard induction with nasal intubation is appropriate. Following loss of consciousness, topical intranasal cocaine may be applied (4% on pledgets, 4 mL maximum) to shrink the nasal mucosa and for vasoconstriction. Side effects are rare but may include ↑ BP, ↑ or ↓ HR, dysrhythmias, and Sz. Other topical vasoconstrictors (e.g., 0.05% oxymetazoline) may be used; however, they are also associated with cardiovascular side effects. The well-lubricated ETT is passed through
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	the nose into the trachea, either blindly or assisted by Magill forceps under direct laryngoscopy. The ETT is often sewn to the nasal septum. The successful use of a flexible reinforced LMA in both adult and pediatric dental patients has been reported. Claimed advantages include no risk of epistaxis and no need for a throat pack, laryngoscopy, or muscle relaxation. Disadvantages include interference with the procedure, throat trauma (e.g., swelling of the epiglottis), and aspiration risk.
ERAS	Verify preincision antibiotics. Maintain normovolemia and normothermia.
Maintenance	Standard maintenance
Emergence	NB: throat packs must be removed prior to extubation. An LMA is typically removed after the patient is awake and able to follow commands.
Blood and fluid requirements	IV: 18 ga × 1 NS/LR @ 4–6 mL/kg/h See Zero Balance Fluid Management, Appendix K, p. K-2 .
Monitoring	Standard monitors
Positioning	✓ pad pressure points ✓ eyes

▼ POSTOPERATIVE

These patients may swallow blood, with consequent PONV. Rx: metoclopramide 10–20 mg iv ± ondansetron 4 mg iv.

Complications	Airway obstruction 2° retained throat pack PONV	Always check for retained throat pack in patients exhibiting symptoms of airway obstruction. Rx: ondansetron 4 mg iv
Multimodal pain management	Oral analgesics	Use nonopioid analgesics. Backup with po opioid

		analgesics
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition and mobilization.

Suggested Readings for Dental/Oral Surgery

1. Al-Jandan BA, Syed FA, Zeidan A, et al: Pharyngeal pack placement in minor oral surgery: a prospective, randomized, controlled study. *Ear Nose Throat J* 2018; 97(3):E18–21.
2. Atan S, Ashley P, Gilthroe MS, et al: Morbidity following dental treatment of children under intubation general anaesthesia in a day-stay unit. *Int J Paediatr Dent* 2004; 14(1):9–16.
3. Cummings DR, Yamashita DD, McAndrews JP: Complications of local anesthesia used in oral and maxillofacial surgery. *Oral Maxillofac Surg Clin North Am* 2011; 23(3):369–77.
4. Hanzawa A, Handa T, Kohkita Y, et al: A comparative study of oral analgesics for postoperative pain after minor oral surgery. *Anesth Prog* 2018; 65(1):24–9.
5. Lin S, McKenna SJ, Yao CF, et al: Effects of hypotensive anesthesia on reducing intraoperative blood loss, duration of operation, and quality of surgical field during orthognathic surgery: a systematic review and meta-analysis of randomized controlled trials. *J Oral Maxillofac Surg* 2017; 75(1):73–86.
6. Mayhew JF: Airway management for oral and maxillofacial surgery. *Int Anesthesiol Clin* 2003; 41(3):57–65.
7. Njuguna B, Delahaye F: Infective endocarditis prophylaxis: a review. *Expert Rev Cardiovasc Ther* 2017; 15(2):127–36.
8. Ogle OE, Mahjoubi G: Local anesthesia: agents, techniques, and complications. *Dent Clin North Am* 2012; 56(1):133–48.
9. Smith MH, Lung KE: Nerve injuries after dental injection: a review of the literature. *J Can Dent Assoc* 2006; 72(6):559–64.
10. Todd DW: Anesthetic considerations for the obese and morbidly obese oral and maxillofacial surgery patient. *J Oral Maxillofac Surg* 2005; 63(9):1348–53.
11. Wakita R, Kohase H, Fukayama H: A comparison of dexmedetomidine sedation with and without midazolam for dental implant surgery. *Anesth Prog* 2012; 59(2):62–8.
12. Yasny JS: Perioperative dental considerations for the anesthesiologist. *Anesth Analg* 2009; 108(5):1564–73.
13. Yu SH, Beirne OR: Laryngeal mask airways have a lower risk of airway complications compared with endotracheal intubation: a systematic review. *J Oral Maxillofac Surg* 2010; 68(10):2359–76.